Maxwell A. Fine

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Education

PhD in Physics & Astrophysics — McGill University (Current)
MSc in Astronomy & Astrophysics — University of Amsterdam (2025)
HBSc in Physics & Astrophysics — University of Toronto (2023)

Summary

Astrophysics PhD building low-latency, high-throughput data pipelines (TB-scale) and conducting large-scale statistical inference. Strong background in Python, Bayesian modeling, and time-series analysis for production systems.

Technical Skills

- **Languages:** Python, Bash, Julia
- **Numerical & ML:** NumPy, Numba, SciPy, Pandas, PyTorch, Scikit-learn, TensorFlow
- **Statistics & Modeling:** Bayesian inference, time-series analysis, stochastic modeling, signal processing, clustering, convolutional neural networks (CNNs)
- **Systems & DevOps:** Linux, Git, Github, Docker, AWS, Kubernetes, Slurm
- **Data & Pipelines:** TB-scale data processing, real-time data processing pipelines, performance-sensitive systems

Selected Publications

- **Fine, M.A.** et al. (2023). Correcting Bandwidth Depolarization by Extreme Faraday Rotation. *MNRAS*. arXiv
- Ould-Boukattine, O.S. et al. (2025). Local environment of repeating FRB 20240619D. *Submitted to MNRAS*. arXiv
- Shin, K. et al. (2025). Discovery of FRB 20240114A. *Submitted to AAS*. arXiv

Awards

- McGill Wolfe Fellowship in Scientific & Technological Literacy — $6,000 (2025)
- ASTRON Summer Graduate Research Fellowship — €2,500 + Housing (2024)
- UofT Summer Undergraduate Research Fellowship — $30,000 (2020–2022)
- NSERC Undergraduate Student Research Award — $6,000 (2021)
- UofT John Pounder Prize in Astronomy — 2x recipient $600 (2019, 2021)

Experience

PhD Researcher — Sept 2025 — Present, *McGill*
CHIME/FRB Collaboration

- Lead studies of statistical analysis using Bayesian inference and time-series modeling, 1 first-author paper, and 3 co-author papers.
- Built and maintain real-time exposure/sensitivity pipeline for calibration and system uptime monitoring.
- Supervised Interns on their data analysis, pipeline development projects.

Graduate Summer Research Fellow — Summer 2024, *Astron*

- Built real-time Python pipeline processing $\sim$1 Gb/s streaming data using clustering and CNN-based detection.
- System deployed on operational radio telescope; detected FRB 20240619D. Project GitHub Repository.

Research Intern — 2022 — 2023
University of Toronto

- Developed Python-based pipeline for multi-messenger search of X-ray/gamma-ray counterparts to FRBs.
- Performed statistical modeling of event fluence using `XSPEC` and `HEAsoft`.

Research Intern (2x) — 2020 — 2021
University of Toronto

- Created a novel RM synthesis algorithm for sources with extreme bandwidth depolarization, contributing to the open-source `RM-Tools` Python package. Used by the Juno space-mission.
- Improved error analysis pipeline for the POSSUM survey, fixing underestimated polarization uncertainties.